

WASP-140b

R-2075

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-140_241116 · created 2026-08-09T23:38:17+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460630.7979 +/- 0.0054 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.167 +/- 0.041
Transit depth (Rp/Rs)^2	2.79 +/- 1.37 [%]
Orbital Inclination (inc)	83.16 +/- 1.76
Airmass coefficient 1 (a1)	1.0 +/- 0.016
Airmass coefficient 2 (a2)	0.004 +/- 0.025
Scatter in the residuals of the lightcurve fit is	1.62 %
Best Comparison Star	#1 - [194, 271]
Optimal Aperture	2.82
Optimal Annulus	10.88
Transit Duration (day)	0.052 +/- 0.032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.167 ± 0.041 vs 0.1432 (deviation 0.58σ)
Duration fit vs published	0.052 d vs 0.0604 d (deviation 0.26σ)
Mid-time O–C	5.7 min
Depth SNR	4.1
Residual scatter	1.62 %

Light curve

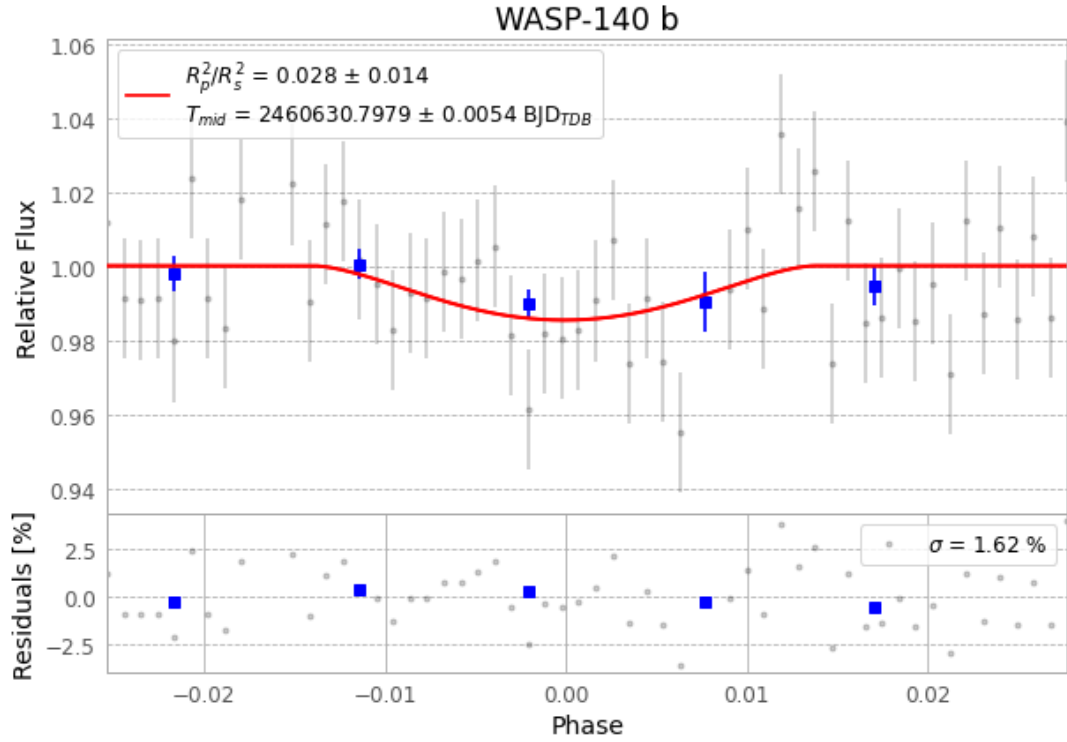


Figure 1 — FinalLightCurve_WASP-140 b_2024-11-15.png (SHA-256 26fbfaebfb14b7a6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [364, 224], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2a9815507102c58c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

56 FITS frames (37 MB), WASP-140241116053916.FITS → WASP-140241116083015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-140-241116.log (883724aac85f...), FinalLightCurve_WASP-140 b_2024-11-15.png (26fbfaebfb14...), FinalLightCurve_WASP-140 b_2024-11-15.csv (b516b5761a98...)

Compute resources

Wall time 137.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 23:38Z.